



```
=====
3. READING DATA FROM WEB
=====
```

First 5 rows of Web data:

	Country	Region
0	Algeria	AFRICA
1	Angola	AFRICA
2	Benin	AFRICA
3	Botswana	AFRICA
4	Burkina	AFRICA

```
=====
4. DATASET INFORMATION
=====
```

CSV Dataset Shape:
(10, 13)

Excel Dataset Shape:
(5, 3)

Web Dataset Shape:
(194, 2)

```
=====
5. CHECKING MISSING VALUES
=====
```

CSV Missing Values:

App	0
Category	0
Rating	1
Reviews	0
Size	0
Installs	0
Type	0
Price	0
Content Rating	0
Genres	0
Last Updated	0
Current Ver	0
Android Ver	0

dtype: int64

Excel Missing Values:

Product	0
Price	0
Quantity	1

dtype: int64

Web Missing Values:

Country	0
Region	0

dtype: int64

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```
[1]: # =====
# EXPERIMENT 2(D)
# EXPLORING DESCRIPTIVE ANALYTICS USING THE IRIS DATASET
# =====

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os

print("=" * 70)
print("EXPERIMENT 2(D) - DESCRIPTIVE ANALYTICS USING IRIS DATASET")
print("=" * 70)

# =====
# 1. CREATE / LOAD IRIS DATASET
# =====

# Create the Iris CSV if it does not already exist.
# This gives us the exact file name expected by the manual.

iris_filename = "iris_dataset(2d).csv"

if not os.path.exists(iris_filename):

    iris = sns.load_dataset("iris")

    iris.to_csv(
        iris_filename,
        index=False
    )

    print("\nIris dataset file created:")
    print(iris_filename)

else:
    print("\nIris dataset file already exists:")
    print(iris_filename)

# Load dataset
df = pd.read_csv(iris_filename)

# =====
# 2. DISPLAY BASIC INFORMATION
# =====
```

```

print("\n" + "=" * 70)
print("BASIC INFORMATION")
print("=" * 70)

df.info()

# =====
# 3. DISPLAY FIRST FIVE ROWS
# =====

print("\n" + "=" * 70)
print("FIRST 5 ROWS")
print("=" * 70)

print(df.head())

# =====
# 4. DISPLAY SUMMARY STATISTICS
# =====

print("\n" + "=" * 70)
print("SUMMARY STATISTICS")
print("=" * 70)

print(df.describe())

# =====
# 5. SPECIES COUNT
# =====

print("\n" + "=" * 70)
print("SPECIES COUNT")
print("=" * 70)

print(df["species"].value_counts())

# =====
# 6. CHECK FOR MISSING VALUES
# =====

print("\n" + "=" * 70)
print("MISSING VALUES")
print("=" * 70)

print(df.isnull().sum())

```

```
# =====
# 12. FINAL RESULT
# =====

print("\n" + "=" * 70)
print("RESULT")
print("=" * 70)

print(
    "The experiment successfully demonstrated descriptive "
    "analytics on the Iris dataset using Pandas and Seaborn."
)

print(
    "Feature distributions, species counts, boxplots, and "
    "feature relationships were analyzed successfully."
)

print("=" * 70)
print("EXPERIMENT 2(D) COMPLETED SUCCESSFULLY")
print("=" * 70)
```

```
=====
EXPERIMENT 2(D) - DESCRIPTIVE ANALYTICS USING IRIS DATASET
=====
```

```
Iris dataset file created:
iris_dataset(2d).csv
```

```
=====
BASIC INFORMATION
=====
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	sepal_length	150 non-null	float64
1	sepal_width	150 non-null	float64
2	petal_length	150 non-null	float64
3	petal_width	150 non-null	float64
4	species	150 non-null	object

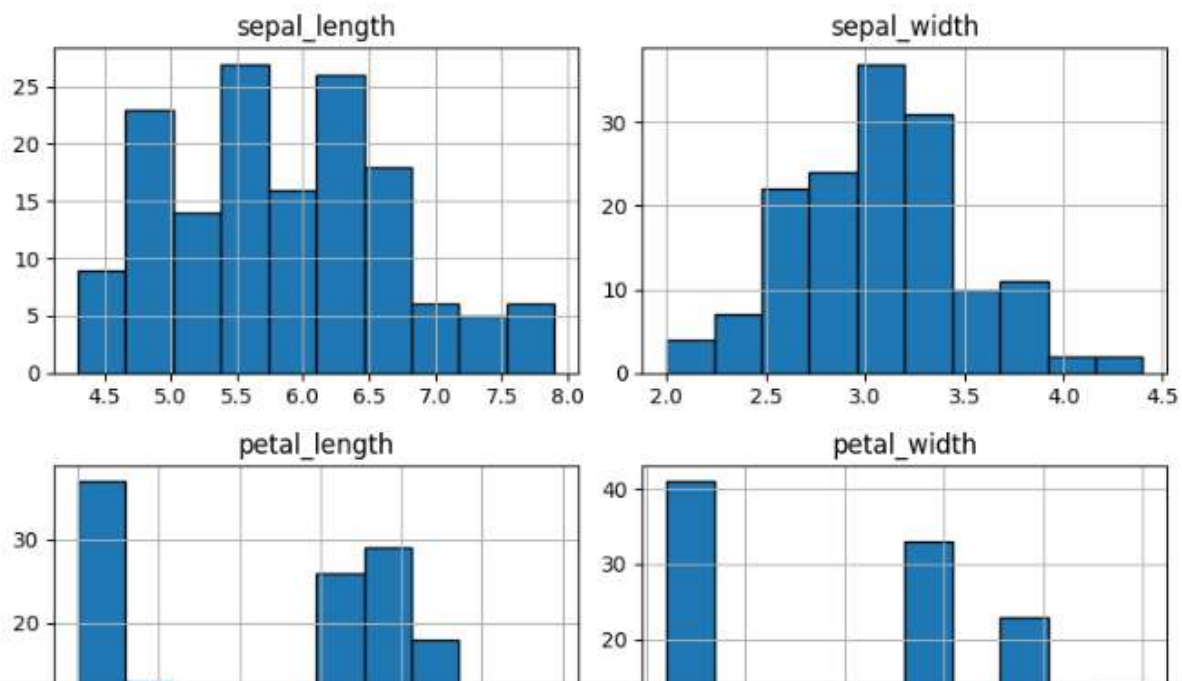
```
dtypes: float64(4), object(1)
```

```
memory usage: 6.0+ KB
```

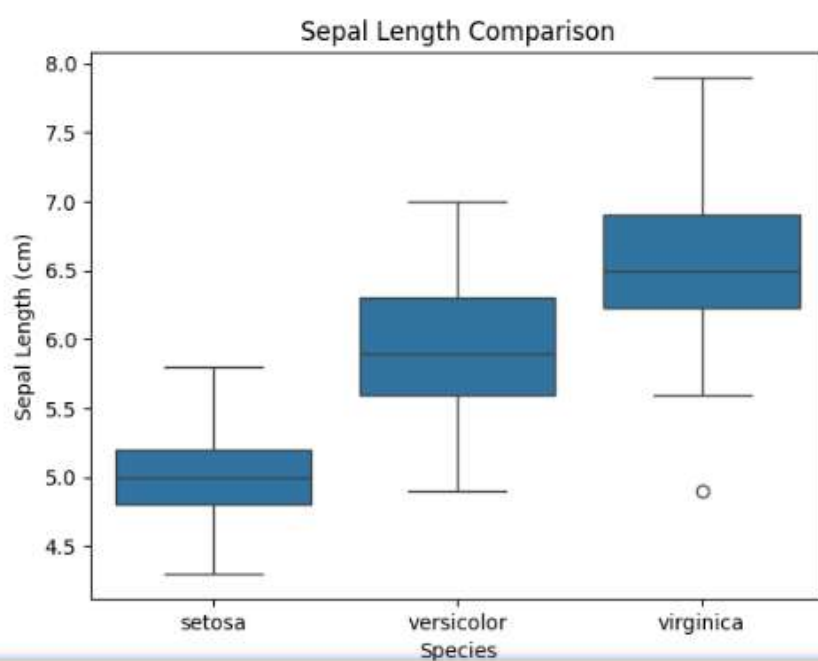
```
=====
FIRST 5 ROWS
=====
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa

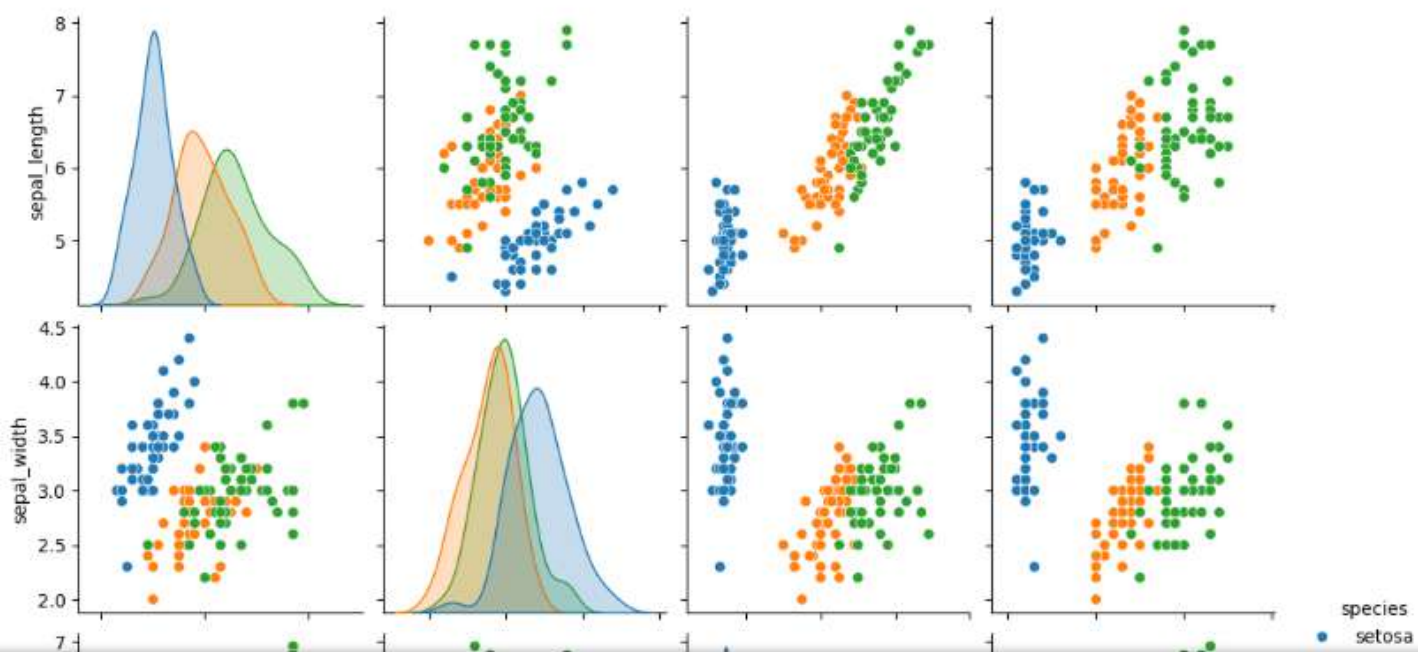
Feature Distributions



BOXPLOT - SEPAL LENGTH



PAIRPLOT - FEATURE RELATIONSHIPS



sepal_length

sepal_width

petal_length

petal_width

=====

MEAN VALUES BY SPECIES

=====

	sepal_length	sepal_width	petal_length	petal_width
species				
setosa	5.006	3.428	1.462	0.246
versicolor	5.936	2.770	4.260	1.326
virginica	6.588	2.974	5.552	2.026

=====

MEDIAN VALUES BY SPECIES

=====

	sepal_length	sepal_width	petal_length	petal_width
species				
setosa	5.0	3.4	1.50	0.2
versicolor	5.9	2.8	4.35	1.3
virginica	6.5	3.0	5.55	2.0

=====

RESULT

=====

The experiment successfully demonstrated descriptive analytics on the Iris dataset using Pandas and Seaborn. Feature distributions, species counts, boxplots, and feature relationships were analyzed successfully.

=====

EXPERIMENT 2(D) COMPLETED SUCCESSFULLY

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